ops-knowledge-loop

BACKGROUND

Prior Art

Three projects that map directly onto this one.

WOLF FRAMEWORK

LangChain · ChromaDB · MCP · local LLM

WOLF Framework

Open-source AI-assisted game dev environment · 2025–present

- LangChain + ChromaDB + LiteLLM RAG via MCP server
- Headless SimBot: evidence-gate-action loop
- Local (Ollama) and cloud/paid variants

Same RAG stack and architectural pattern as this project.

QVC / QURATE RETAIL

~850 services · 5y zero audit breach

Cloud Migration

~850 Kubernetes microservices · 2020–2024

- Migrated off Jenkins, Spinnaker, Rancher, Bitbucket, Ansible fully to Azure DevOps
- Live multi-DC AEM deployments; up to 40 stakeholders
- Zero audit breach over 5 years; ServiceNow tracked

Same retail-scale infrastructure, ServiceNow, and gating discipline.

MULTI-GEO MOBILE PIPELINE

Release: 3h → 30min
PR acceptance: 2h → 20min

Build-Test-Deploy

Bitbucket · Jenkins · Artifactory · Jira · Consul

- Multi-geo mobile build-test-deploy pipeline
- Release effort: 3 hours → 30 minutes
- Automated PR acceptance: 2 hours → 20 minutes
- Stage tracking, drift detection, rollback paths

Same compression pattern applied to incident response.

8+ years automotive & UK retail · Ansible · ServiceNow · multi-geo release engineering · RAG / MCP / local-LLM pipelines

CONTEXT

The On-Call Problem

MEAN TIME TO REMEDIATION

Mean time to remediation is dominated by lookup time, not fix time

KNOWLEDGE SILO

Institutional knowledge lives in engineers' heads, not in systems

REPEAT WORK

Repeated incidents get manually triaged every single time

▸ Ember Grid runs 17 microservices across UK retail infrastructure

ARCHITECTURE

How It Works

[01] INCIDENT
ServiceNow ticket free-text description

→

[02] RAG SEARCH
sentence-transformers + ChromaDB top-3 matches

→

[03] LLM REASONING
Ollama qwen3:14b → structured JSON response

→

[04] DECISION GATE
≥0.70 auto-fire · <0.70 human review

→

[05] OUTCOME
Rundeck job fires or engineer gets pre-reasoned brief

▸ Fully local · No data leaves the machine · No API costs · ~30s end-to-end

DATA

What the RAG Searches

▸ INCIDENT HISTORY

500+ synthetic incidents across 17 services

Each indexed as a vector embedding in ChromaDB

OOM kills · latency spikes · ETL failures · EDI mismatches · POS outages · loyalty bugs · notification duplicates

▸ RUNBOOKS

Per-service markdown runbooks

checkout · product-search · payment-processor · …

Auto-generated runbooks from resolved ServiceNow tickets

Chunked and embedded alongside incident history

All embeddings generated with all-MiniLM-L6-v2 · Similarity threshold: 0.60 strong match

TOOLING

Ops Dashboard

- 📋 Recent incidents and their triage status
- 🎯 RAG match scores per incident
- 🤖 AI recommendation with confidence score
- ⚡ Decision gate outcome (auto-executed vs pending review)
- 🔧 Rundeck job catalogue

`Flask · localhost:5000`

DASHBOARD PREVIEW

```
┌──────────────────────────────┐
│ ops-knowledge-loop dashboard │
├──────┬────────────────┬──────┤
│ INC  │ STATUS         │ CONF │
├──────┼────────────────┼──────┤
│ 0043 │ AUTO-EXECUTE   │ 0.95 ✓ │
│ 0044 │ PENDING        │ 0.35 ▪ │
│ 0045 │ AUTO-EXECUTE   │ 0.75 ✓ │
└──────┴────────────────┴──────┘
```

RESULTS

10 Queries · Live Run

5 AUTO-EXECUTED

5 PENDING REVIEW

gate threshold: 0.70

INCIDENT	RECOMMENDED JOB	CONFIDENCE	OUTCOME
checkout-service OOM kill	restart-service-with-memory-bump	0.95	AUTO-EXECUTE
payment-processor 500 errors	rollback-to-previous-version	0.95	AUTO-EXECUTE
product-search unresponsive	reindex-elasticsearch	0.75	AUTO-EXECUTE
inventory sync stalled	force-inventory-sync	0.66	PENDING REVIEW
store POS tills unresponsive	rollback-to-previous-version	0.75	AUTO-EXECUTE
loyalty service not awarding pts	restart-service-rolling	0.35	PENDING REVIEW
order fulfilment latency spike	scale-up-replicas	0.75	AUTO-EXECUTE
notification service dup emails	restart-service-with-memory-bump	0.35	PENDING REVIEW
supplier EDI orders dropped	force-inventory-sync	0.65	PENDING REVIEW
recommendation engine degraded	reindex-elasticsearch	0.45	PENDING REVIEW

"The gate is discriminating by design — weak RAG similarity or ambiguous root cause correctly suppresses auto-execution"

LIVE OUTPUT — QUERY 1 / 10

checkout-service OOM kill · 0.95 -> AUTO-EXECUTE

```
$ python query_live.py "checkout service OOM kill, container hitting memory limit and restarting repeatedly under load"

┌──────────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬────────────────┬──────────────────────┬──────────────────────────────────────────┐
│ # │ Source         │        Similarity    │ Excerpt                                  │
├───┼────────────────┼──────────────────────┼──────────────────────────────────────────┤
│ 1 │ incidents.json │   0.797 ✓ strong match │ [INC0043096] checkout-service OOM kill   │
│ 2 │ incidents.json │   0.796 ✓ strong match │ [INC0043144] checkout-service OOM kill   │
│ 3 │ incidents.json │   0.795 ✓ strong match │ [INC0043048] checkout-service OOM kill   │
└───┴────────────────┴──────────────────────┴──────────────────────────────────────────┘

✓ Strong precedent found — incidents.json (similarity 0.797)

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : restart-service-with-memory-bump
  Confidence       : 0.95
  Decision Gate    : AUTO-EXECUTE
  AI Source        : ollama:qwen3:14b
  Reasoning        : OOM kills due to memory limits directly align with the job purpose.
                     Runbook context explicitly confirms memory threshold breach as trigger.

┌──────────────────────────────────────────────────────────────────────────────┐
│ ✓ WOULD AUTO-EXECUTE  —  restart-service-with-memory-bump  (conf 0.95 ≥ 0.70 gate)  │
└──────────────────────────────────────────────────────────────────────────────┘
Total runtime: 8.7s
```

LIVE OUTPUT — QUERY 3 / 10

product-search unresponsive after deploy · 0.75 -> AUTO-EXECUTE

```
$ python query_live.py "product search returning empty results, search index unresponsive after deployment"

┌──────────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬──────────────────┬──────────────────────────┬──────────────────────────────────────┐
│ # │ Source           │        Similarity        │ Excerpt                              │
├───┼──────────────────┼──────────────────────────┼──────────────────────────────────────┤
│ 1 │ product-search.md│  0.575 ~ partial match   │ Service Overview: ES-backed search service   │
│ 2 │ incidents.json   │  0.571 ~ partial match   │ [INC0043346] product-search slow query 1352ms│
│ 3 │ product-search.md│  0.570 ~ partial match   │ Diagnostic Steps: kubectl exec cache rate    │
└───┴──────────────────┴──────────────────────────┴──────────────────────────────────────┘

⚠ No close precedent — best match 0.575 in product-search.md. AI will deduce a remediation.

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : reindex-elasticsearch
  Confidence       : 0.75
  Decision Gate    : ABOVE GATE → AUTO-EXECUTE
  AI Source        : ollama:qwen3:14b
  Reasoning        : Unresponsive ES index after deployment maps to reindexing as direct fix.
                     Runbook + incident context both confirm the Elasticsearch-backed architecture.

┌──────────────────────────────────────────────────────────────────────────────┐
│ ✓ WOULD AUTO-EXECUTE  —  reindex-elasticsearch  (confidence 0.75 ≥ 0.70 gate)       │
└──────────────────────────────────────────────────────────────────────────────┘
Total runtime: 9.9s
```

LIVE OUTPUT — QUERY 6 / 10

loyalty service not awarding points · 0.35 -> PENDING REVIEW

```
$ python query_live.py "loyalty service not awarding points after purchase, customer accounts not updating"

┌──────────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬────────────────┬──────────────────────┬──────────────────────────────────────────┐
│ # │ Source         │        Similarity    │ Excerpt                                  │
├───┼────────────────┼──────────────────────┼──────────────────────────────────────────┤
│ 1 │ incidents.json │  0.642 ~ partial match │ [INC0042145] loyalty-service failing to award pts│
│ 2 │ incidents.json │  0.562 ~ partial match │ [INC0041652] loyalty-service reward API 500      │
│ 3 │ incidents.json │  0.561 ~ partial match │ [INC0041134] loyalty-service negative balances   │
└───┴────────────────┴──────────────────────┴──────────────────────────────────────────┘

⚠ No close precedent — best match 0.642. AI will deduce a remediation.

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : restart-service-rolling
  Confidence       : 0.35  ← MODEL SELF-DOWNGRADED
  Decision Gate    : RULE-BASED FALLBACK
  AI Source        : ollama:qwen3:14b
  Reasoning        : Past loyalty-service issues resolved via restarts or config updates.
                     Runbook context references backfill scripts — not a restart path.
                     Semantic mismatch between query and available context. Self-downgraded.

┌──────────────────────────────────────────────────────────────────────────────┐
│ ▪ PENDING HUMAN REVIEW  —  restart-service-rolling  (conf 0.35 < 0.70 gate)  │
└──────────────────────────────────────────────────────────────────────────────┘

Total runtime: 14.6s
```

DESIGN PHILOSOPHY

The Gate is the Point

loyalty-service · 0.35
RAG matched past incidents but runbook referenced a backfill script, not a restart. Model self-downgraded.

notification-service · 0.35
Matched incident was push notifications, not email. Semantic mismatch caught.

supplier-EDI · 0.65
Schema validation failure; force-inventory-sync is plausible but not certain. Correctly held for review.

```
$ python query_live.py "loyalty service not awarding points after purchase, customer accounts not updating"

Step 1 — Searching knowledge base...
┌───┬────────────────┬──────────────────────┬──────────────────────────────────────────────┐
│ # │ Source         │       Similarity     │ Excerpt                                      │
├───┼────────────────┼──────────────────────┼──────────────────────────────────────────────┤
│ 1 │ incidents.json │  0.642 ~ partial match │ [INC0042145] loyalty-service failing to award pts  │
│   │                │                      │ for click-and-collect — missing order type mapping  │
│ 2 │ incidents.json │  0.562 ~ partial match │ [INC0041652] reward redemption API 500 (DB pool)   │
│ 3 │ incidents.json │  0.561 ~ partial match │ [INC0041134] points returning negatives (bulk event) │
└───┴────────────────┴──────────────────────┴──────────────────────────────────────────────┘

⚠ No close precedent — best match 0.642. AI will deduce a remediation.

  Recommended Job  : restart-service-rolling
  Confidence       : 0.35  ← MODEL SELF-DOWNGRADED
  Decision Gate    : RULE-BASED FALLBACK
  Reasoning        : Runbook context references a backfill script — not a restart path.
                     Semantic mismatch between query and available context.

■  PENDING HUMAN REVIEW  —  restart-service-rolling (confidence 0.35 < 0.70 gate)
   On-call engineer sees this recommendation + RAG sources before any action is taken.
Total runtime: 14.6s
```

"A system that knows what it doesn't know is more useful than one that always fires"

Observability

Log Every Step
RAG top-K + similarity scores, LLM token counts + latency, every gate confidence and outcome, every Rundeck execution result.

Dashboards
Confidence distribution over time, auto-execution rate, false positive rate, and P99 latency per pipeline stage.

Alerts
Confidence distribution shift, fallback rate increase, and any Rundeck failure without a subsequent human resolution.

Sudden confidence drops are a production smell: stale knowledge base, model version drift, or a retrieval-quality regression.

TECHNOLOGY

Stack

Python 3.11
Core pipeline

ChromaDB
Vector store

sentence-transformers
all-MiniLM-L6-v2 embeddings

Ollama + qwen3:14b
Local LLM · 9.3GB · Q4_K_M · RTX 4090

Flask
Ops dashboard

Rich
CLI output formatting

ServiceNow API
Mocked ticket source

Rundeck
Mocked job execution

▸ Runs entirely on local hardware · Zero external API calls · Zero ongoing cost

ROADMAP

Where This Goes Next

Shadow Deployment Pipeline

A shadow environment mirrors live incidents without acting. Once recommendations earn sufficient confidence, a single approval promotes to production - zero-downtime, the system validates itself before it acts.

Repository-Aware Context

Continuous indexing of service repos gives the LLM code-level context - config files, dependency changes, recent commits - when a root cause needs investigation beyond tickets and runbooks.

Living Organisational Memory

Automatic ingestion of JIRA, Confluence, and equivalent knowledge bases keeps RAG context current with the team's accumulated decisions and post-mortems. The system learns from every resolved incident.

LIVE OUTPUT — QUERY 2 / 10

payment-processor 500 errors · 0.95 -> AUTO-EXECUTE

```
$ python query_live.py "payment processor returning 500 errors, transactions not completing at checkout"

┌──────────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬────────────────┬──────────────────────┬──────────────────────────────────────────┐
│ # │ Source         │        Similarity    │ Excerpt                                  │
├───┼────────────────┼──────────────────────┼──────────────────────────────────────────┤
│ 1 │ incidents.json │    0.495 ✗ weak match │ [INC0043135] payment-processor 5xx spike (70%)  │
│ 2 │ incidents.json │    0.492 ✗ weak match │ [INC0043231] payment-processor 5xx spike (72%)  │
│ 3 │ incidents.json │    0.488 ✗ weak match │ [INC0043183] payment-processor 5xx spike (71%)  │
└───┴────────────────┴──────────────────────┴──────────────────────────────────────────┘

⚠ No close precedent — best match 0.495. AI will deduce a remediation.

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : rollback-to-previous-version
  Confidence       : 0.95
  Decision Gate    : AUTO-EXECUTE
  AI Source        : ollama:qwen3:14b
  Reasoning        : Three prior payment-processor 5xx spikes all resolved by rollback.
                     Standard mitigation confirmed despite weak RAG similarity scores.

┌──────────────────────────────────────────────────────────────────────────────┐
│ ✓ WOULD AUTO-EXECUTE  —  rollback-to-previous-version  (conf 0.95 ≥ 0.70 gate)  │
└──────────────────────────────────────────────────────────────────────────────┘
Total runtime: 8.2s
```

inventory sync stalled · 0.66 -> PENDING REVIEW

```
$ python query_live.py "inventory sync stalled, stock levels on website not updating after warehouse batch job"

┌──────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬────────────────┬──────────────────────┬──────────────────────────────────────────┐
│ # │ Source         │        Similarity    │ Excerpt                                  │
├───┼────────────────┼──────────────────────┼──────────────────────────────────────────┤
│ 1 │ incidents.json │  0.675 ~ partial match │ [INC0041047] inventory-sync batch job failed     │
│ 2 │ incidents.json │  0.656 ~ partial match │ [INC0043477] inventory-sync pipeline stall 75min │
│ 3 │ incidents.json │  0.639 ~ partial match │ [INC0042043] inventory-sync failing 14 stores    │
└───┴────────────────┴──────────────────────┴──────────────────────────────────────────┘

✓ Strong precedent found — incidents.json (similarity 0.675)

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : force-inventory-sync
  Confidence       : 0.66
  Decision Gate    : PENDING REVIEW
  AI Source        : ollama:qwen3:14b
  Reasoning        : Stalled sync mirrors INC0043477 pipeline-stall scenario exactly.
                     force-inventory-sync would resolve — confidence just below gate threshold.

┌──────────────────────────────────────────────────────────────────────────────┐
│ ▪ PENDING HUMAN REVIEW  —  force-inventory-sync  (confidence 0.66 < 0.70 gate)     |
└──────────────────────────────────────────────────────────────────────────────┘
Total runtime: 15.4s
```

LIVE OUTPUT — QUERY 5 / 10

store POS tills unresponsive · 0.75 -> AUTO-EXECUTE

```
$ python query_live.py "store POS system tills unresponsive, cashiers cannot process customer transactions"

┌──────────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬────────────────┬──────────────────────┬──────────────────────────────────────────┐
│ # │ Source         │        Similarity    │ Excerpt                                  │
├───┼────────────────┼──────────────────────┼──────────────────────────────────────────┤
│ 1 │ incidents.json │  0.573 ~ partial match │ [INC0043165] store-pos-system pipeline stall  │
│ 2 │ incidents.json │  0.569 ~ partial match │ [INC0043333] store-pos-system pipeline stall  │
│ 3 │ incidents.json │    0.547 ✗ weak match  │ [INC0041062] store-pos-system API latency spike │
└───┴────────────────┴──────────────────────┴──────────────────────────────────────────┘

⚠ No close precedent — best match 0.573. AI will deduce a remediation.

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : rollback-to-previous-version
  Confidence       : 0.75
  Decision Gate    : ABOVE GATE → AUTO-EXECUTE
  AI Source        : ollama:qwen3:14b
  Reasoning        : P1 incident — Consul mesh misconfiguration caused 2,300-store outage.
                     Configuration rollback was the confirmed resolution. Pattern matches.

┌──────────────────────────────────────────────────────────────────────────┐
│ ✓ WOULD AUTO-EXECUTE  —  rollback-to-previous-version  (conf 0.75 ≥ 0.70 gate)  │
└──────────────────────────────────────────────────────────────────────────┘
Total runtime: 9.9s
```

LIVE OUTPUT — QUERY 7 / 10

order fulfilment latency spike · 0.75 -> AUTO-EXECUTE

```
$ python query_live.py "order fulfilment delays caused by upstream pricing engine latency spike during peak hours"

┌──────────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬────────────────┬──────────────────────┬──────────────────────────────────────────┐
│ # │ Source         │        Similarity    │ Excerpt                                  │
├───┼────────────────┼──────────────────────┼──────────────────────────────────────────┤
│ 1 │ incidents.json │  0.571 ~ partial match │ [INC0043496] order-fulfilment P99 breach (1102ms)│
│ 2 │ incidents.json │  0.565 ~ partial match │ [INC0043448] order-fulfilment P99 breach (4926ms)│
│ 3 │ incidents.json │  0.558 ~ partial match │ [INC0043304] order-fulfilment P99 breach (798ms) │
└───┴────────────────┴──────────────────────┴──────────────────────────────────────────┘

⚠ No close precedent — best match 0.571. AI will deduce a remediation.

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : scale-up-replicas
  Confidence       : 0.75
  Decision Gate    : ABOVE GATE → AUTO-EXECUTE
  AI Source        : ollama:qwen3:14b
  Reasoning        : Upstream latency causes downstream throughput pressure.
                     Scaling replicas reduces per-instance load while root cause is addressed.

┌──────────────────────────────────────────────────────────────────────────┐
│ ✓ WOULD AUTO-EXECUTE  —  scale-up-replicas  (confidence 0.75 ≥ 0.70 gate)  │
└──────────────────────────────────────────────────────────────────────────┘
Total runtime: 12.2s
```

LIVE OUTPUT — QUERY 8 / 10

notification service duplicate emails · 0.35 -> PENDING REVIEW

```
$ python query_live.py "notification service sending duplicate confirmation emails to customers after every order event"

┌──────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬──────────────────────┬──────────────────────┬──────────────────────────────────────┐
│ # │ Source               │        Similarity    │ Excerpt                              │
├───┼──────────────────────┼──────────────────────┼──────────────────────────────────────┤
│ 1 │ incidents.json       │  0.687 ~ partial match │ [INC0041294] notification-svc duplicate  │
│   │                      │                      │ PUSH notifications (order-fulfilment retry)│
│ 2 │ incidents.json       │    0.520 ✗ weak match │ [INC0041158] notification-svc email queue  │
│ 3 │ INC0042468-notif-...md │   0.467 ✗ weak match │ Auto-generated runbook: unsubscribe links  │
└───┴──────────────────────┴──────────────────────┴──────────────────────────────────────┘

✓ Strong precedent found — incidents.json (similarity 0.687)
  [note: matched push notifications, not email — semantic mismatch]

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : restart-service-with-memory-bump
  Confidence       : 0.35  ← MODEL SELF-DOWNGRADED
  Decision Gate    : RULE-BASED FALLBACK
  AI Source        : ollama:qwen3:14b
  Reasoning        : Runbook context (INC0042468) used restart-with-memory-bump for config issue.
                     Current incident involves email duplicates, not push notifications.
                     Semantic mismatch between incident type and available context.

┌──────────────────────────────────────────────────────────────────────────────────┐
│ ▪ PENDING HUMAN REVIEW  —  restart-service-with-memory-bump  (conf 0.35 < 0.70)   │
└──────────────────────────────────────────────────────────────────────────────────┘
Total runtime: 22.5s
```

supplier EDI orders dropped · 0.65 -> PENDING REVIEW

```
$ python query_live.py "supplier integration service dropping EDI purchase orders during XML schema validation"

┌──────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬────────────────┬──────────────────────┬──────────────────────────────────────────┐
│ # │ Source         │        Similarity    │ Excerpt                                  │
├───┼────────────────┼──────────────────────┼──────────────────────────────────────────┤
│ 1 │ incidents.json │  0.705 ~ partial match │ [INC0041737] supplier-integration EDI batch  │
│   │                │                      │ failure — Supplier-31 schema version mismatch │
│ 2 │ incidents.json │  0.561 ~ partial match │ [INC0043383] supplier-integration ETL failure │
│ 3 │ incidents.json │    0.542 ✗ weak match  │ [INC0041195] supplier-integration EDI timeout │
└───┴────────────────┴──────────────────────┴──────────────────────────────────────────┘

✓ Strong precedent found — incidents.json (similarity 0.705)

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : force-inventory-sync
  Confidence       : 0.65
  Decision Gate    : PENDING REVIEW
  AI Source        : ollama:qwen3:14b
  Reasoning        : EDI orders dropped during XML schema validation matches INC0041737 closely.
                     force-inventory-sync addresses data pipeline stalls, not schema mismatches.
                     Plausible mitigation, but not the correct root-cause fix. Correctly held.

┌──────────────────────────────────────────────────────────────────────────┐
│ ▪ PENDING HUMAN REVIEW  —  force-inventory-sync  (confidence 0.65 < 0.70 gate)  │
└──────────────────────────────────────────────────────────────────────────┘
Total runtime: 25.7s
```

LIVE OUTPUT — QUERY 10 / 10

recommendation engine degraded · 0.45 -> PENDING REVIEW

```
$ python query_live.py "recommendation engine returning identical product suggestions for all users after overnight model retraining"

┌──────────────────────────────────────┐
│ ops-knowledge-loop  |  Live Incident Query │  MOCK_MODE=TRUE  |  env: mock
└──────────────────────────────────────┘

Step 1 — Searching knowledge base...
┌───┬────────────────┬──────────────────────┬──────────────────────────────────────────┐
│ # │ Source         │        Similarity    │ Excerpt                                  │
├───┼────────────────┼──────────────────────┼──────────────────────────────────────────┤
│ 1 │ incidents.json │    0.393 ✗ weak match │ [INC0041839] product-search autocomplete empty   │
│ 2 │ incidents.json │    0.356 ✗ weak match │ [INC0043065] product-search feature flag misconf │
│ 3 │ incidents.json │    0.315 ✗ weak match │ [INC0043017] product-search feature flag misconf │
└───┴────────────────┴──────────────────────┴──────────────────────────────────────────┘

⚠ No close precedent — best match 0.393. AI will deduce a remediation.
  [all three matches are from a different service; zero relevant knowledge base entries]

Step 2 — Requesting AI remediation recommendation...
  Recommended Job  : reindex-elasticsearch
  Confidence       : 0.45  ← ACKNOWLEDGING LIMITS
  Decision Gate    : RULE-BASED FALLBACK
  AI Source        : ollama:qwen3:14b
  Reasoning        : Identical suggestions indicate stale/improperly updated index data.
                     No recommendation-engine incidents in knowledge base. Cross-service inference
                     only. Model self-downgraded — this is the system acknowledging its limits.

┌──────────────────────────────────────────────────────────────────────────────┐
│ ▪ PENDING HUMAN REVIEW  —  reindex-elasticsearch  (confidence 0.45 < 0.70 gate)   │
└──────────────────────────────────────────────────────────────────────────────┘
Total runtime: 13.8s
```